

Korean Petrochemical Industry

Decarbonization Pathway Analysis

Marginal Abatement Cost Curve (MACC) Model

Report Generated: 2025-10-12 00:39

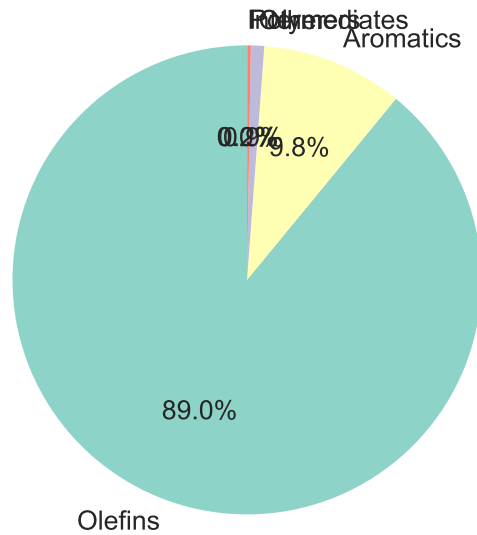
Model Version: 2.2

Key Model Parameters

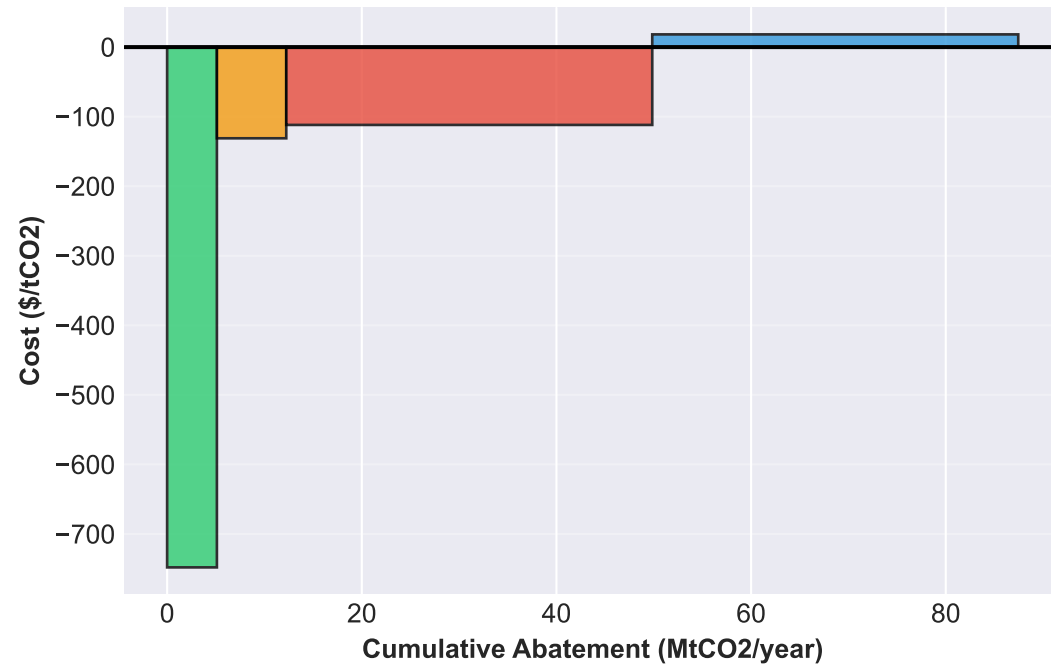
- Baseline Emissions (2025): 52.0 MtCO₂/year
 - Facilities Analyzed: 248
- Technologies: 4 (Heat Pump, RE PPA, NCC-H₂, NCC-Electricity)
 - Time Horizon: 2025-2050
- Maximum Abatement Potential: 49.8 MtCO₂ (95.9%)
 - Scenarios: Conservative, Moderate, Aggressive

Executive Summary

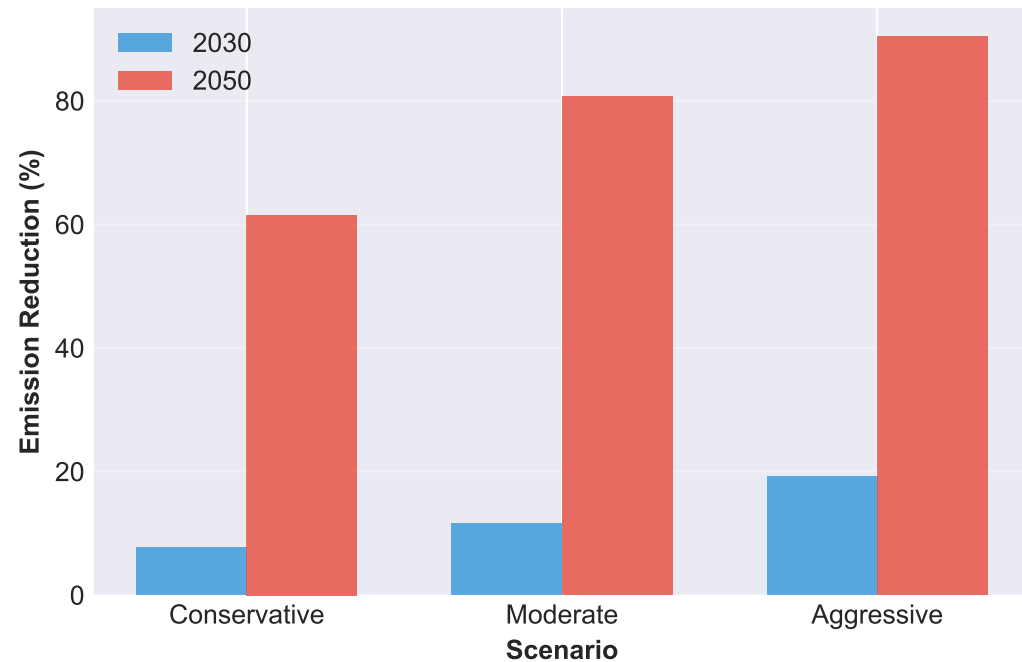
2025 Baseline by Product Group (52.0 MtCO₂ total)



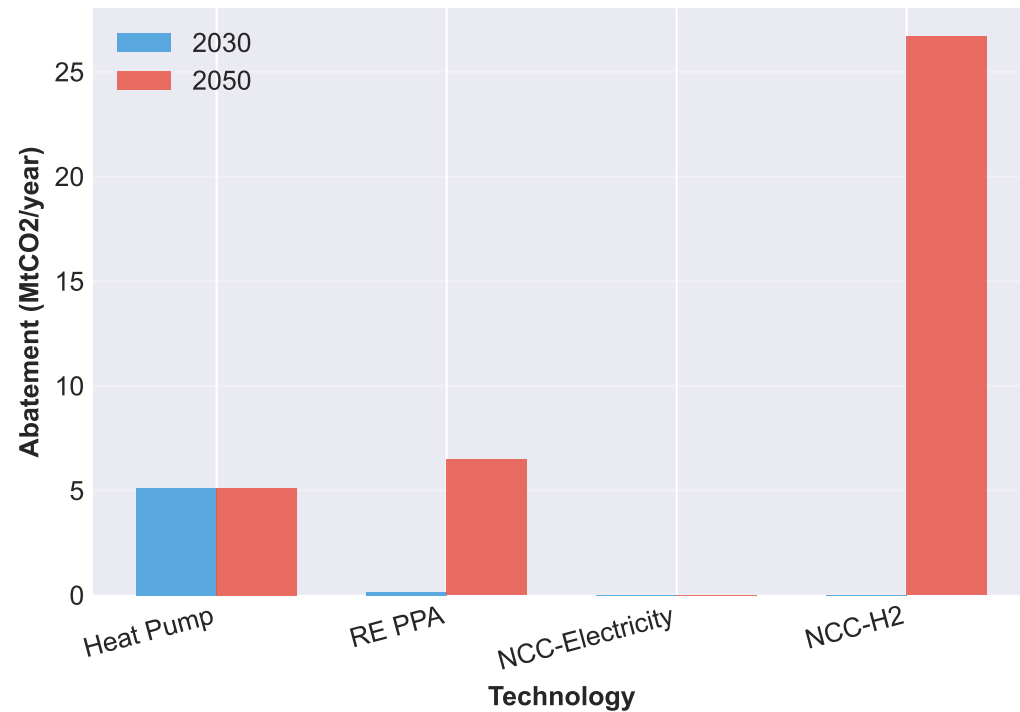
MACC Curve - 2030 (3 cost-negative technologies)



Emission Reduction by Scenario

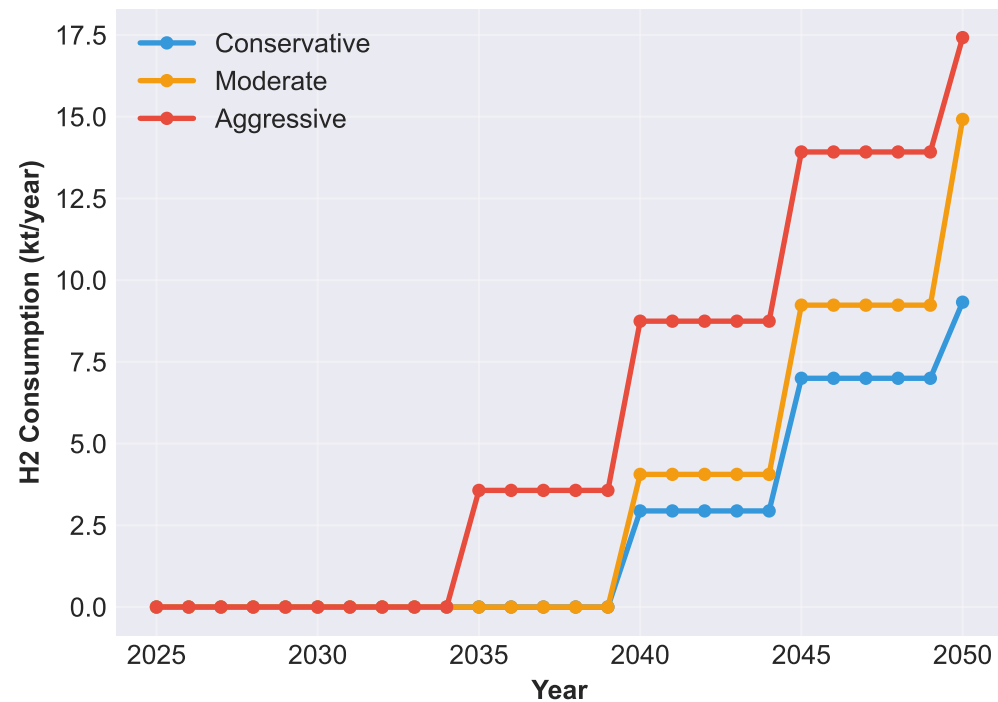


Technology Deployment - Moderate Scenario

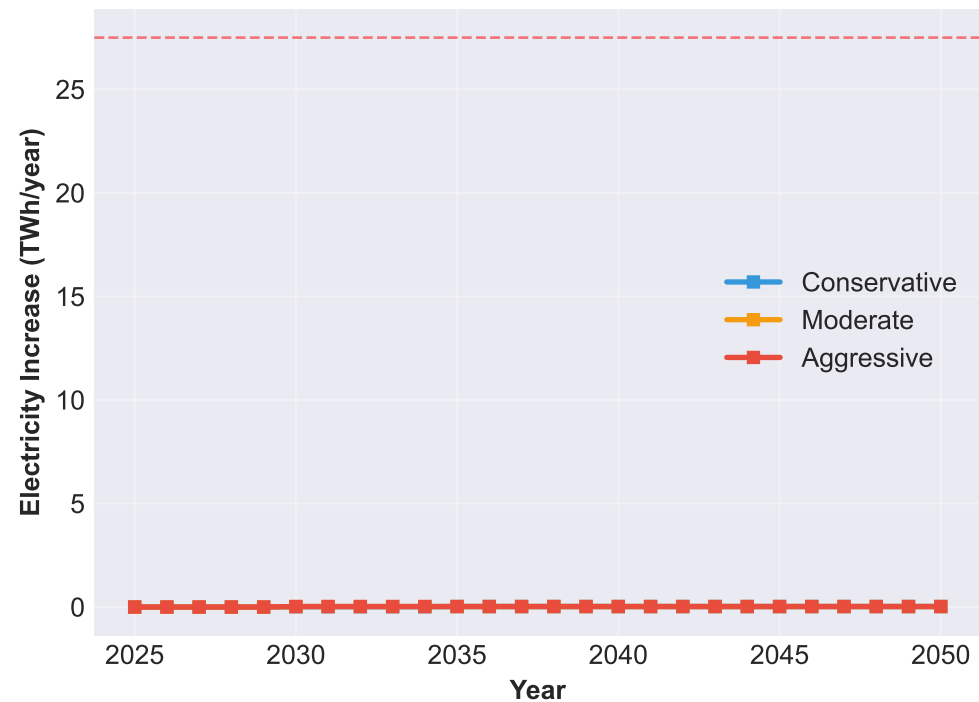


Energy Demand & Investment Analysis

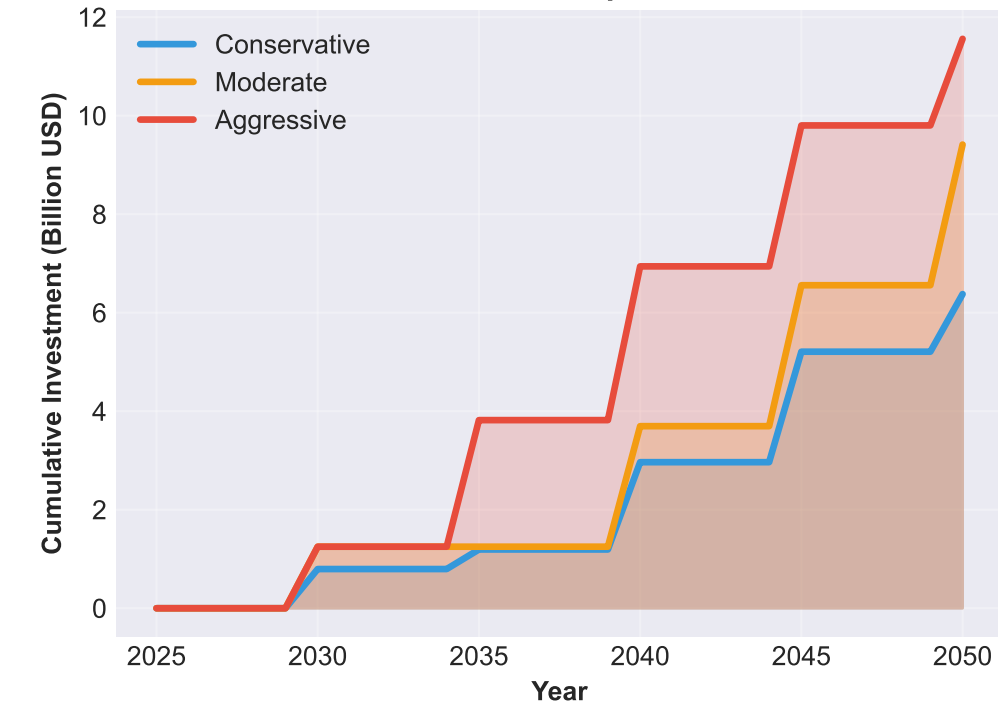
Hydrogen Demand Trajectory



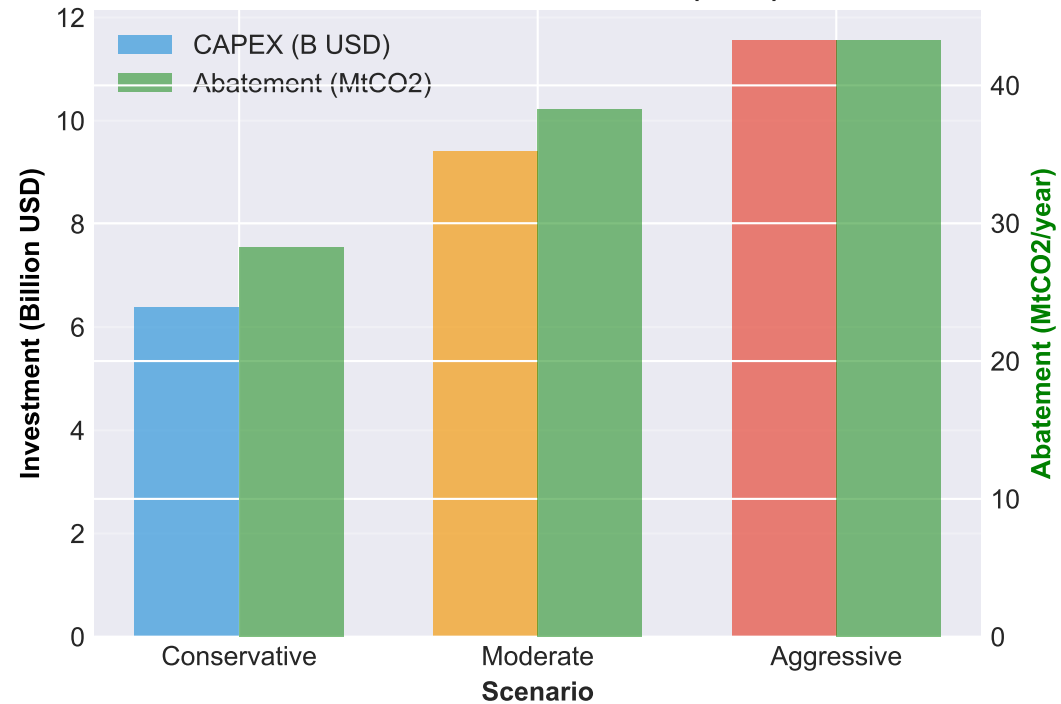
Additional Electricity Demand



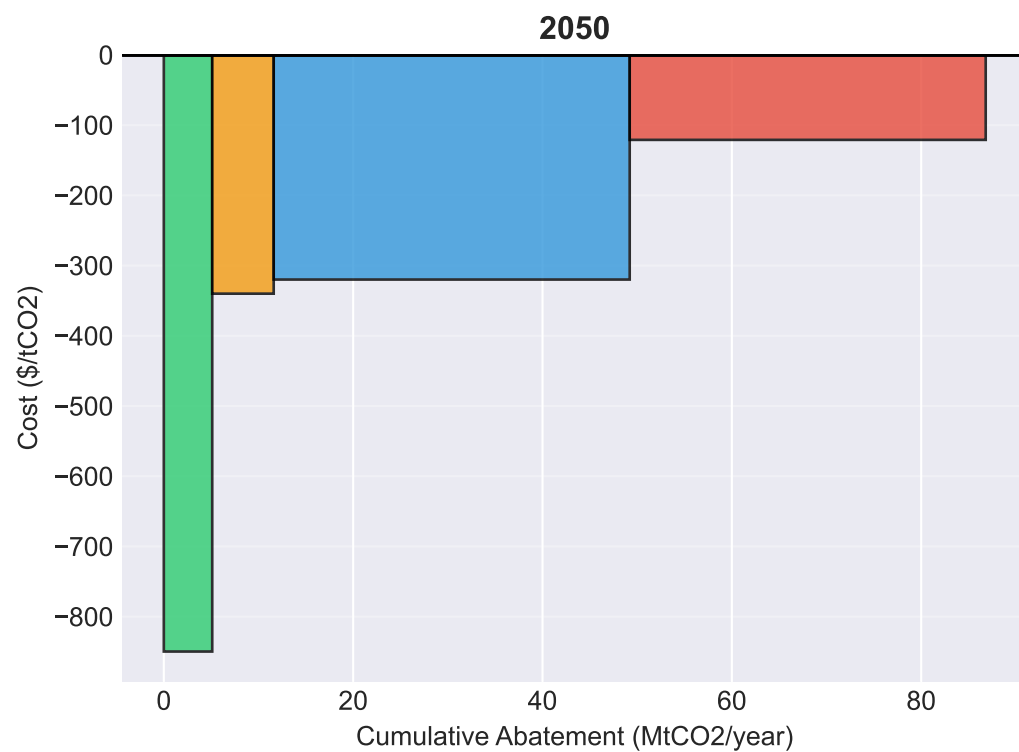
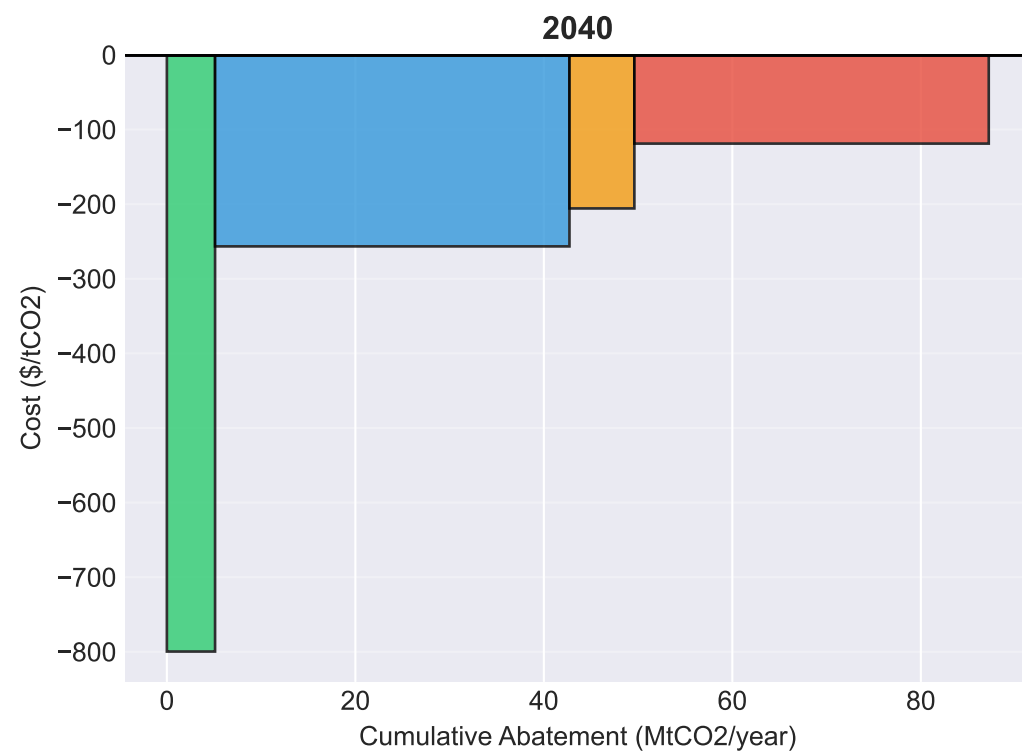
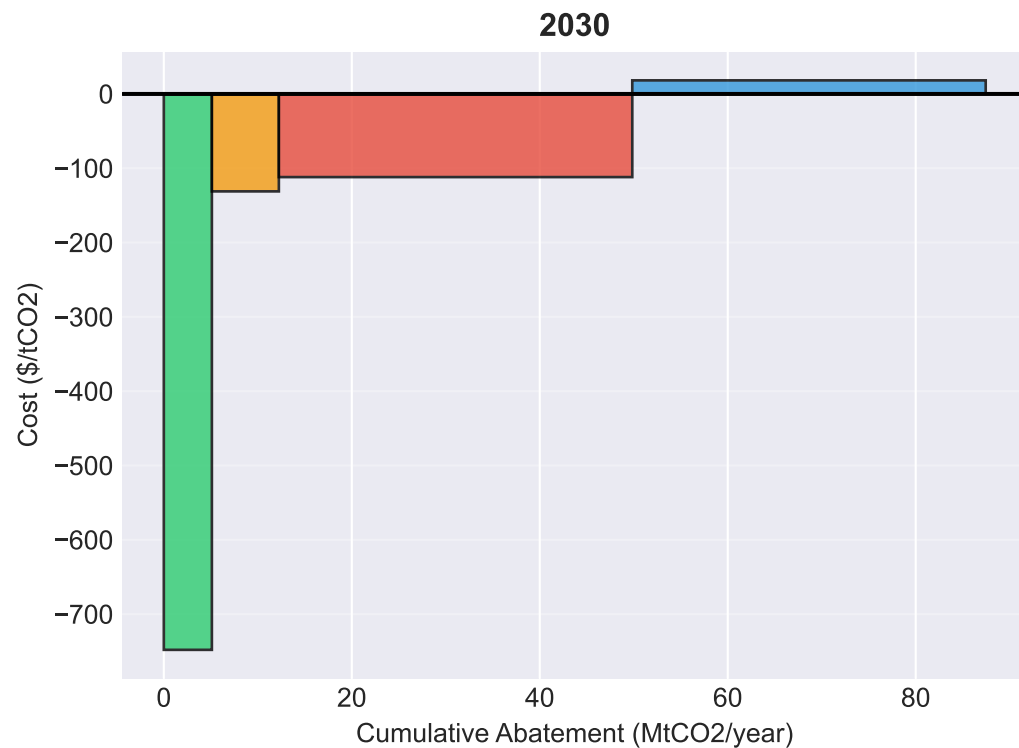
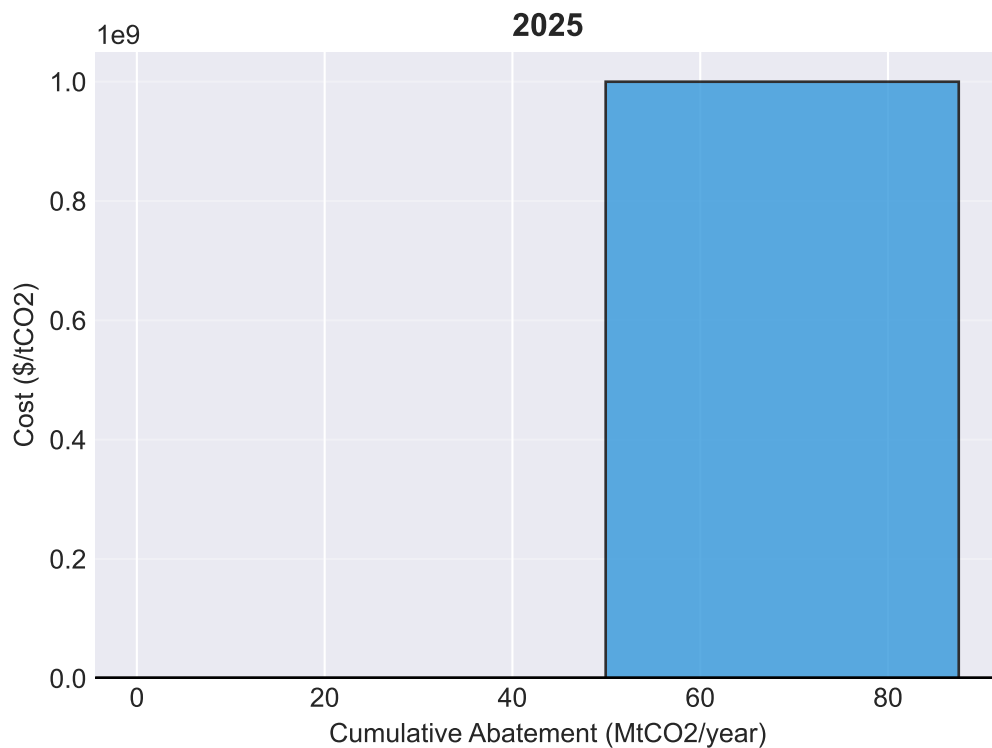
Total CAPEX Requirements



Investment vs Abatement (2050)

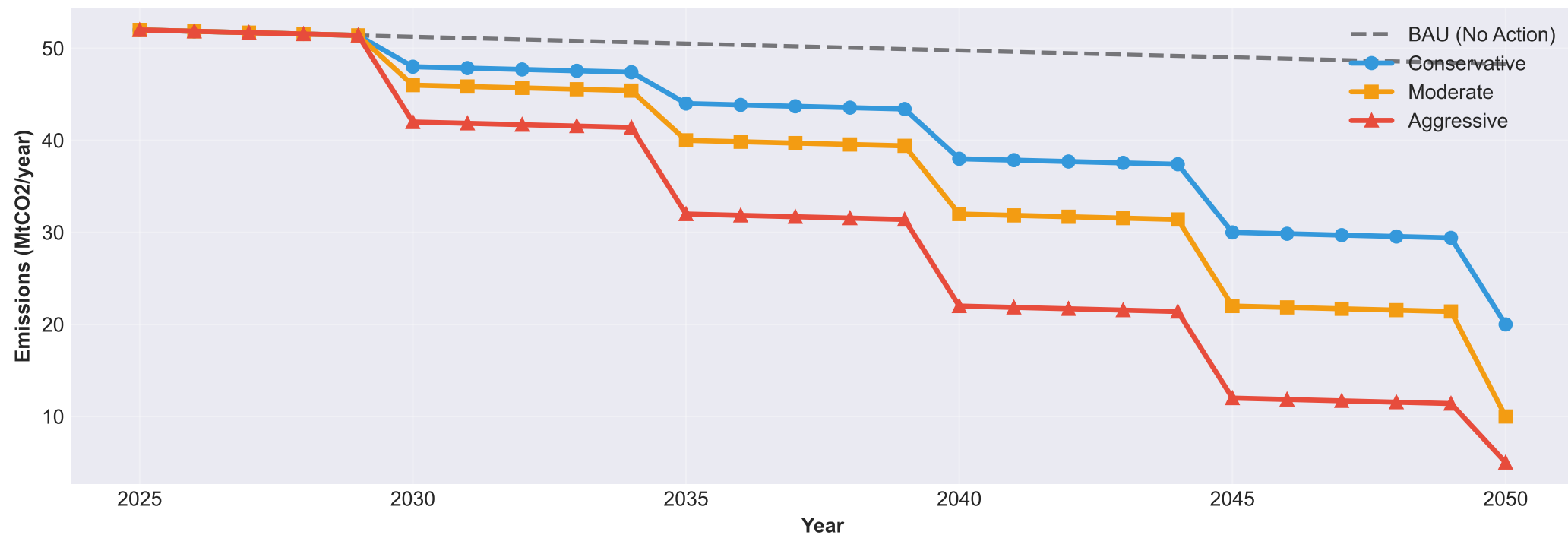


MACC Curve Evolution (2025-2050)

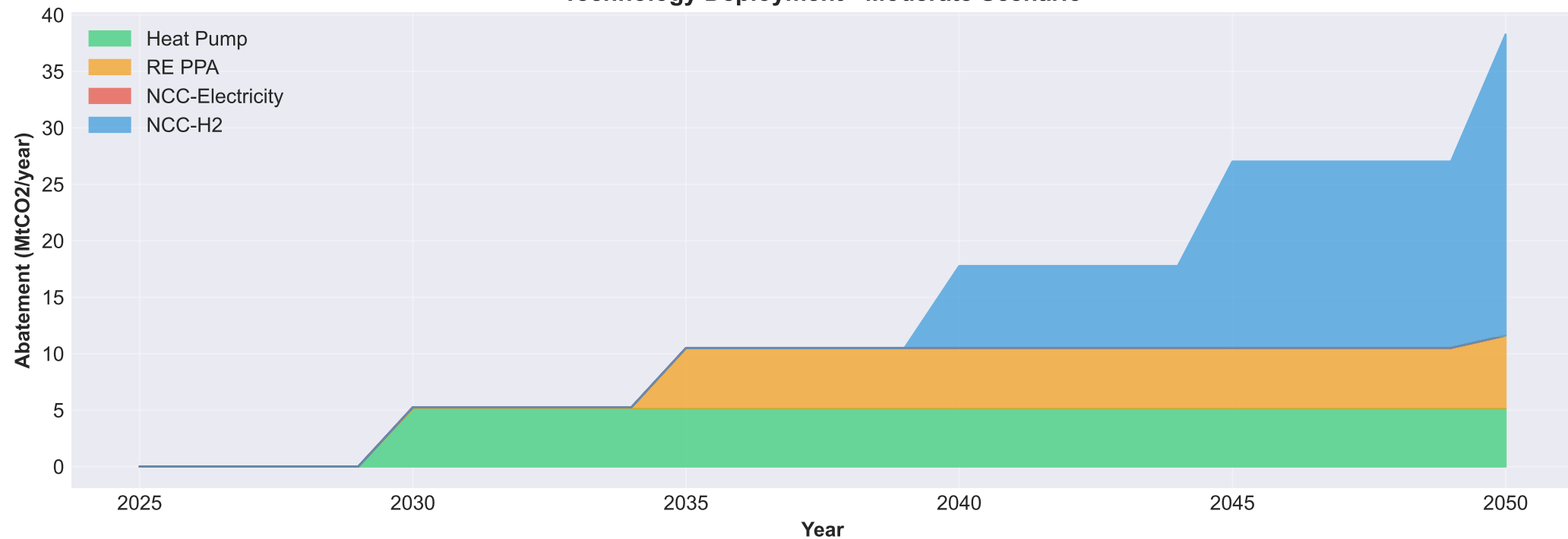


Emission Trajectories by Scenario

Annual Emissions



Technology Deployment - Moderate Scenario



Summary Tables

Scenario Comparison

Scenario	2030 Emis (Mt)	2030 Reduct (%)	2050 Emis (Mt)	2050 Reduct (%)
Conservative	48.0	7.7%	20.0	61.5%
Moderate	46.0	11.5%	10.0	80.8%
Aggressive	42.0	19.2%	5.0	90.4%

Technology Costs - 2030

Technology	Abatement (MtCO2)	CAPEX+OPEX (\$/tCO2)	Fuel Diff (\$/tCO2)	Total Cost (\$/tCO2)
Heat_Pump	5.11	\$12.6	\$-760.6	\$-748.0
NCC-H2	37.60	\$24.0	\$0.0	\$18.3
NCC-Electricity	37.60	\$28.7	\$0.0	\$-111.9
RE_PPA	7.13	\$0.0	\$-131.1	\$-131.1